

DOINGG@UNION.EDU

COURSE: CSC106 CAN COMPUTERS THINK?

SELF

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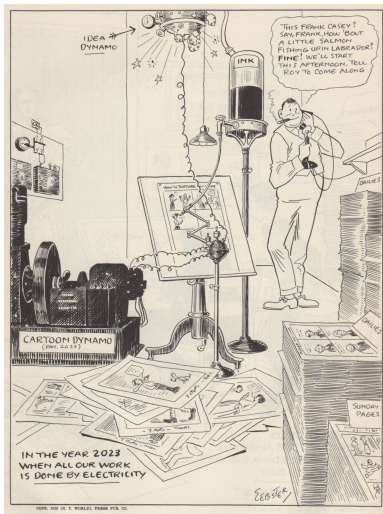
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The course website for [CSC106](#), part of the Union College [CS curriculum](#). Here you can find our weekly schedule, assignments and resources.

Announcements

Office Hours

- Student/Office Hours:
 - Steinmetz 217
 - Monday 4:00 pm - 5:30 pm

- Thursday 3:00 pm – 4:30 pm
- subject to change, check course website for most up-to-date schedule
- drop-in or schedule a 15 minute slot: <https://calendar.app.google/8bus6pfDvyphR9ar5>
- by appointment for another time or over zoom

Course Description

Can computers think? Is it possible to build intelligent machines? What does it mean for a machine to think or to be intelligent? Why, or why not, would we want intelligent machines? To what extent do we already have them? How do they work? How do they impact our lives? What are their limitations? This course invites you to explore these questions and more.

This course is an introduction to the field of computer science with an artificial intelligence theme that does not assume you have prior experience with computer programming or computer science. It introduces basic algorithms, data structures and programming techniques. You will learn how computer scientists think about and solve problems by doing your own computer programming. This will give you an understanding of how the current technology for building intelligent machines works. In addition, you will read non-technical papers about artificial intelligence and the ethical questions raised by its use. You will reflect on the ways current computational systems often reinforce biases and can harm groups of people inequitably. And you will evaluate how the choices you are making when creating your own programs will affect future users.

If you're thinking about a CS or CPE major or minor, this course will give you a solid foundation to continue to *Programming on Purpose* (CSC 120), to *Data Structures* (CSC 151) and beyond. If you're a neuroscience major, it will help you see how scientists are using biological and neurological principles to model "behavior" in computers and it will teach you skills useful for processing experimental data. If you are thinking about another major or minor, this course will give you foundations to apply computing in whatever area you are interested in. And if you're here just because you're curious, that's great!

Schedule

Weekly Schedule

Schedule

check often as things may change

W	TOPIC	Due by end of week	Runestone
1	Programming, Syntax and Semantics	Portfolio (PF) 1	1 & 2
2	Variables and Functions	PF 2	3 & 4
3	Conditional Statements	PF 3, Project Demo 1	5
4	Conditional Loops	PF 4, Practical Exam 1	6
5	Counting Loops	PF 5, Project Demo 2	7
6	Strings, Lists	PF 6, Written Exam 1	8 & 9
7	Nested Lists, Dictionaries	PF 7, Project Demo 3	10
8	File I/O	PF 8, Practical Exam 2	11
9	Recursion	PF 9, Project Demo 4	
10	Searching, Sorting	PF 10	

W	TOPIC	Due by end of week	Runestone
11	Finals Week Monday 11/23 2:30 - 4:30	Written Exam 2	

Grading

Table 2: Grading Scheme

Course Element	Grade Point Contribution	Notes
Reading & Practice	5%	Runestone Interactive Textbook
Portfolios & Share-outs	15%	From in-class and take-home problems
Programming	20%	4 at 5% each
Project Demos		
Practical Exams	20%	2 at 10% each
Midterm	20%	
Final Exam	20%	written
total	100%	

Table 3: Letter Grade Scale

Letter Grade	Percent range
A	93 - 100
A-	90 - 92
B+	87 - 89
B	83 - 86
B-	80 - 82
C+	77 - 79

Letter Grade	Percent range
C	73 - 76
C-	70 - 72
D	60 - 69
F	0 - 59

Note: You must get a C- or better in order to take any other class that has a CSC-10x prerequisite. Several larger programming projects will be assigned to reinforce the concepts discussed in class and put into practice in homework assignments. All projects can be completed using programming techniques that have been covered in class.

Do not use techniques that have not been covered in class unless specifically told otherwise. Part of the challenge for each programming project is figuring out how to complete the assignment using only the tools provided in the course up to that point.

Assignments

HW 01

CSC106: Week 1

- due Monday 9/14/26 by class time (1:50 pm)
- [Intro Survey](#)
- Runestone Chapters 1 & 2
- Finish all parts of the Week 1 Portfolio
 - especially note parts we don't start in class and you will have to do on your own ("oyo")
 - there may also be parts we planned to start in class but run out of time for... regardless of the planned in-class start and share-out times, you are responsible for finishing the portfolio by the due date
- Do your share-out
 - check the schedule to see what problem and what date you will share:
 - add your solution to the shared slide-deck:

Weekly Notes

Week 1

Notes from Week 1

- [Seymour Pappert Turtle Documentary](#)
 - Turtle drawing activity
- [Instructions for First Program Activity](#)
- [Reeborg activity: newspaper](#)

Python Examples

- check back here for in-class activities

About

References

